

Lab 7

Objective: Practice Pointers and Dynamic Arrays (Chapter 9).

Instructions: finish the sample given below which practice the following:

1. Dynamically allocate an array of integers of the given size and populate the array with numbers.
2. Write a function `double calculateAverage(int* arr, int size)` that takes a pointer to an array and its size as arguments and the function calculate and return the average of the array elements.
3. Create a function `int* resizeArray(int* arr, int oldSize, int newSize)` that doubles the size of an existing dynamic array and transfers the existing data.

```
1  #include <iostream>
2  using namespace std;
3
4  //function that takes a pointer to an array and its size as arguments,
5  //and return the average of the array elements.
6  double calculateAverage(int* arr, int size) {
7      int sum = 0;
8      for (int i = 0; i < size; i++) {
9          sum += arr[i];
10     }
11     return static_cast<double>(sum) / size;
12 }
13 int* resizeArray(int* arr, int oldSize, int newSize) {
14     int* newArray = new int[newSize];
15     for (int i = 0; i < oldSize; i++) {
16         newArray[i] = arr[i];
17     }
18     delete[] arr; // Free old array memory
19     return newArray;
20 }
21 int main() {
22
23     //Prompt the user for the size of an array.
24     int size;
25     cout << "Enter the size of the array: ";
26     cin >> size;
27
28     //Dynamically allocate an array of integers of the given size.
29     int* array = new int[size]; // Dynamic array allocation
30 }
```

```

31 // Populate and print the array
32 for (int i = 0; i < size; i++) {
33     array[i] = i + 1;
34     cout << "array[" << i << "] = " << array[i] << endl;
35 }
36 //use calculateAverage function
37 double avg = calculateAverage(array, size);
38 cout << "Average of array elements: " << avg << endl;
39 //use resizeArray function
40 int newSize = size * 2;
41 array = resizeArray(array, size, newSize);
42
43 cout << "Resized array: ";
44 for (int i = 0; i < newSize; i++) {
45     cout << array[i] << " ";
46 }
47 cout << endl;
48 delete[] array; // Free memory
49 return 0;
50 }

```

Submission:

You need to upload to Moodle: the source code (.cpp files), and the screenshot of the running of your program before 4pm Wednesday.